25900288

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSE) - MG3DSECHE202 – Safe Laboratory Practices In Chemistry
(2024 ADMISSION ONWARDS)

Instruction to the Examiners:

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.

Part A

Short Answer Type Questions. Answer any 5 Questions. Each Question Carries 4 Marks

1) What are the key components of an effective emergency response plan?

- Risk Assessment and Preparedness**:** Identify potential hazards, assess risks, and develop prevention and mitigation strategies.
- Clear Roles and Responsibilities**:** Define duties of emergency response team members and establish a clear chain of command.
- Communication Procedures: Ensure reliable communication channels for internal coordination and external alerts (e.g., emergency services, public).
- Training and Drills: Conduct regular training, simulations, and reviews to ensure staff readiness and plan effectiveness. **[Any 3 relevant points related to this answer give 4 Mark]**

2) Explain briefly the safe handling of toxic compounds.

Safe handling of toxic compounds involves a combination of **precautionary practices, protective equipment, and proper procedures** to minimize exposure and environmental contamination. Key points include:1). **Risk Assessment, 2). Personal Protective Equipment (PPE) 3). Engineering Controls 4).Safe Work Practices 5). Spill and Waste Management** 6). .**Training and Documentation**

[Any 3 relevant points related to this answer give 4 Mark]

3) Why laboratory equipment safety is important?

Proper use of laboratory equipment helps avoid burns, cuts, electric shocks, and chemical spills that can cause serious harm to students and staff. Safe handling prevents breakage or malfunction of expensive and delicate instruments, ensuring they remain in good working condition for future use. Following safety procedures avoids contamination and errors during experiments, leading to correct and dependable scientific results. Safety practices promote discipline, cleanliness, and responsibility, creating a secure environment for effective learning and research.

[Any 3 relevant points- Give 4 Mark]

4) Explain the role of fire extinguishers.

The role of fire extinguishers is to control or put out small fires before they spread and cause major damage or injury. They are an essential part of emergency safety equipment in laboratories, workplaces, and public areas

Fire control: Used to extinguish or suppress fires at an early stage., Safety protection**:** Helps protect people, property, and equipment from fire hazards, Emergency response: Provides a quick and effective first response before firefighters arrive. Different types for different fires: Designed for specific classes of fires (e.g., electrical, flammable liquids, ordinary combustibles).

[Any 3 relevant points related to this answer give 4 Mark]

5) Explain why glassware is preferred in laboratory experiments compared to other materials

Glassware is preferred in laboratory experiments because it possesses several important properties that make it suitable for accurate and safe scientific work. Glass does not react with most acids, bases, or solvents, ensuring that experiments are not contaminated by reactions with the container. It can withstand high and low temperatures, allowing it to be used for heating, cooling, or boiling substances.

The clear nature of glass allows easy observation of reactions, color changes, and measurements. Glass can be easily sterilized and reused without affecting experimental results. Prevents absorption of chemicals, maintaining purity of substances used.

[Any 3 relevant points related to this answer give 4 Mark]

6) Explain the toxic effect of picric acid

Picric acid is 2,4,6-trinitrophenol and its Toxic Effects are **Skin and Eye Irritation**:, **Respiratory Effects**, **Gastrointestinal Effects**, **Systemic Toxicity**: Absorbed picric acid can affect the liver, kidneys, and central nervous system, causing symptoms such as headache, dizziness, fatigue, and confusion, **Blood Effects:** May cause methemoglobinemia—a condition where hemoglobin cannot effectively carry oxygen, leading to cyanosis (bluish skin)

[Any 3 relevant points related to this answer give 4 Mark]

7) Explain the significance of fire safety symbols.

Fire safety symbols enable people to instantly recognize potential fire risks and the presence of flammable or combustible materials. These symbols indicate the locations of fire extinguishers, alarms, fire exits, and emergency assembly points, helping people act quickly and safely. By alerting individuals to dangers and reminding them to follow proper safety procedures, fire safety symbols help in reducing the chances of fire outbreaks. Fire safety symbols use standardized colors, shapes, and pictograms that are easily understood by everyone, regardless of language or literacy level, ensuring effective safety communication.

[Any 3 relevant points- Give 4 Mark]

Part B

Short Essay Type Questions, Answer any 5 questions. Each question carries 7 marks

8) Why must oxidizing and reducing agents be stored separately? Illustrate with an example?

Oxidizing and reducing agents must be stored separately because they can react violently with each other, often causing fire, explosion, or release of toxic gases even without any external ignition source. Eg. Potassium Permaganate is a strong oxidizing agent and Glycerol a reducing agent. If these two come in contact, they ignite spontaneously due to an intense redox reaction. Immediate combustion without any spark or flame.

[Give full credit to reasonable Explanation-3.5 mark and any one example 3.5 Mark]

9) Give a short essay on safe work procedures for handling hazardous chemicals.

All chemical containers must be clearly labeled with the chemical name, concentration, hazard symbols, and date of preparation. Use of Personal Protective Equipment (PPE) [gloves, goggles, lab coats, face shield and respirators. Fume hoods are necessary to prevent inhalation of toxic vapors or dust. Chemicals must be stored separately according to their properties—for example, oxidizing and reducing agents, acids and bases, should not be kept together and Flammable substances must be kept away from open flames or heat sources. Chemicals must be handled with clean, dry hands and appropriate tools such as tongs or pipettes. Spills should be cleaned immediately using proper neutralizing agents. Chemical waste must be disposed of according to approved procedures, never poured into sinks or drains. Safety equipment such as fire extinguishers, eye wash stations, and safety showers should be easily accessible. Workers must be trained to respond quickly in case of spills, fires, or chemical exposure

[Any Reasonable five points - Give full credit]

10) Discuss the general rules for disposal of chemical waste

1. Identify and Classify Waste. **2. Segregate Waste Properly:** Different types of waste should never be mixed. For example: Acids and bases must be separated,**3. Label All Waste Containers:** Each waste container should be clearly labelled with the chemical name(s), concentration, hazard symbols, and date. **4. Use Suitable Containers:** Store chemical waste in containers made of compatible materials—for example, glass bottles for acids **5. Temporary Storage and Handling:** Chemical waste should be kept in designated storage areas, away from incompatible materials and sources of ignition. Storage areas must be well ventilated **6. Disposal Through Authorized Channels: 7. Record Keeping: 8. Emergency Preparedness:** Spill kits, absorbent materials, and personal protective equipment (PPE) should be readily available in all waste storage and handling areas. **(Any Reasonable 5 points give full marks)**

11) Compare the functions of respirators and masks in providing safety to laboratory workers

Respirators protect the wearer from inhaling hazardous substances like chemicals, dust, and biological agents. **Masks** mainly prevent the spread of droplets from the wearer to the environment. **Respirators** provide a tight seal around the face for maximum protection. **Masks** fit loosely and offer limited filtration. **Respirators** use filters or cartridges to remove harmful particles and gases. **Masks** are suitable for basic hygiene and low-risk environments. **Respirators** require maintenance and proper fitting, while **masks** are usually disposable or washable for simple use**.[Any three comparison give 7 marks]**

12) Explain the importance of using hazard pictograms in chemical labeling, with examples

a)Immediate Hazard Recognition: Pictograms allow anyone handling a chemical—scientists, students, or workers—to instantly recognize potential hazards without needing to read detailed text **b)**Supports Safe Handling and Storage: Each pictogram indicates specific precautions (e.g., wear gloves, avoid inhalation, use fume hood).c)Prevents Health Hazards: Some hazards (like carcinogenicity or respiratory sensitization) are not immediately visible. Pictograms alert users even when the danger isn't obvious.d) Legal and Regulatory Compliance: Use of correct pictograms is **required by law** under systems like **OSHA Hazard Communication Standard (HCS)** e)**Universal Understanding**- Pictograms are **language-independent**, ensuring that people from different linguistic backgrounds can recognize the

same hazard.

Pictogram	Symbol	Example Substance	Hazard Type
☠️ Skull and Crossbones	Acute toxicity	Sodium cyanide	Poisonous
🔥 Flame	Flammable	Ethanol	Fire hazard
⚗️ Corrosion	Corrosive	Sulfuric acid	Burns metals/skin
💥 Exploding Bomb	Explosive	TNT	Explosion hazard
🩺 Health Hazard (silhouette with star)	Chronic health hazard	Benzene	Carcinogenic
❗ Exclamation Mark	Irritant	Ammonia solution	Causes irritation
🌿 Environment	Environmental hazard	Pesticides	Toxic to aquatic life
🔥 Over Circle	Oxidizer	Hydrogen peroxide	Intensifies fire
🫧 Gas Cylinder	Gas under pressure	Propane	Compressed gas risk

(Explanation – 4 marks and give any 2 examples- 3 Marks)

13) How do we respond to chemical spill?

Immediately inform others nearby and, if the spill is large or involves toxic fumes, evacuate the area**.** Identify the type and quantity of chemical spilled. Only trained personnel should handle hazardous or unknown chemicals. Wear appropriate gloves, goggles, lab coat, and **mask** before attempting cleanup to avoid contact or inhalation. Avoid letting the chemical enter drains. Neutralize acids or bases if possible (e.g., use sodium bicarbonate for acid spills). Collect the waste in a labeled disposal container. Wash the area thoroughly with water or suitable cleaning agents and ensure good ventilation to remove fumes. Notify the safety officer or supervisor and record the spill details for future prevention. **[Any reasonable 5 points - Give full credit]**

14) Explain the significance of waste containers in maintaining safety during laboratory and industrial waste disposal.

Waste containers play a **vital role in ensuring safety** during the handling, storage, and disposal of waste generated in laboratories and industries.. The following points explain their significance:1. **Segregation of Waste:** Different types of waste—chemical, biological, radioactive, and general—must be kept separate to prevent dangerous chemical reactions or contamination. 2.**Containment of Hazardous Materials:** Waste containers are designed to safely hold hazardous materials, preventing leaks, spills, and the release of toxic vapors. 3.**Prevention of Accidents and Injuries:** Using proper containers—such as sharps containers for needles and broken glass—prevents injuries and infections caused by accidental contact with sharp or contaminated objects.4.**Environmental Protection:** Secure waste containers prevent the release of harmful chemicals and biological agents into air, soil, or water, thereby protecting the ecosystem from pollution and contamination.5.**Compliance with Regulations:** Proper labeling, storage, and disposal using approved waste containers ensure compliance with environmental, health, and safety regulations set by authorities. 6.**Ease of Handling and Transport:** Well-designed containers make it easier to store, transport, and dispose of waste safely.7. **Minimization of Cross-Contamination:** Keeping waste in designated containers reduces the risk of cross-contamination between hazardous and non-hazardous materials. **(Any 5 Reasonable points- Give Full Mark)**

Part C

Essay Type Questions, Answer any 1 questions. Each question carries 15 marks

15) What systemic failures led to the Bhopal Gas Disaster, and how can the lessons learned be applied to enhance modern industrial safety and accountability?

Safety systems were not properly maintained. The refrigeration unit (to keep MIC below 5°C) was shut down to save money. Safety devices (Gas Scrubber, alarm system) not working .Over-storage of MIC. No proper containment system or emergency vent to control gas leaks.

Modern industrial safety and accountability

The Bhopal tragedy taught the world that safety, transparency, and accountability must come before profit. Regular monitoring, trained personnel, and strict enforcement of safety standards are essential to prevent such disasters.

Enforce **strong laws** for hazardous industries. Regular **inspection and monitoring** by safety authorities. Ensure all safety devices (alarms, scrubbers, flare towers, etc.) are working and regularly checked. Prevent **cost-cutting** at the expense of safety. Workers must receive **proper training** to handle dangerous chemicals. Regular **safety drills and emergency response training** should be mandatory. Companies must **share information** about hazardous materials and risks with employees and local authorities. Maintain **clear labeling and documentation** of chemicals. Every plant should have a **disaster management plan** and **emergency evacuation system**. Hospitals and

local bodies should be **trained to respond** to chemical accidents. Hazardous industries should be built **away from residential areas**. Companies must be **legally responsible** for safety failures. Install automatic monitoring systems and real-time leak detectors to identify hazards early. **(Reasonable Explanation for systemic failures (4 points)-7.5 marks and reasonable explanation for lessons learned (4 points) -7.5 marks)**

16) Explain the significance of Hazard (H) and Precautionary (P) statements in chemical safety

Hazard statement is assigned a code beginning with "H" followed by three digits (e.g., H200, H315, H400). First number designating the type of hazard as follows:"2" for physical hazards,"3" for health hazards,"4" for environmental hazards.

Significance**: 1.Describe the type and severity of chemical hazards:** H-statements clearly indicate whether a chemical is flammable, toxic, corrosive, carcinogenic, or environmentally harmful.2. **Facilitate quick risk assessment:** Standardized codes (e.g., H225 – highly flammable liquid and vapor) help workers identify risks immediately.3. **Promote consistent communication:** Globally recognized wording reduces confusion across different languages and regions.4.**Guide safe handling and storage:** Knowledge of hazards allows selection of appropriate **personal protective equipment (PPE)** and preventive measures.

Precautionary Statements (P-codes): Precautionary statements describe recommended measures to minimize or prevent adverse effects from exposure, improper handling, or storage of chemicals. Its main purpose is to provide practical safety advice for users — how to protect themselves and the environment. Each precautionary statement is assigned a code beginning with "P" followed by three digits (e.g., P102, P280, P301).

Significance: 1.**Provide instructions for safety measures:** P-statements advise on prevention, response, storage, and disposal of chemicals.2. **Reduce accidents and exposure:** For example, *P210: Keep away from heat/sparks/open flames*.3.**Ensure emergency preparedness:** Statements provide steps for first aid, spill response, and containment.4. **Support regulatory compliance:** Following P-statements aligns with occupational safety and environmental regulations. **(Explanation for Hazard (H) (4 points) – 7.5 marks and Precautionary (P) (4 points) - 7.5 marks)**
